Onishi Tatsuki
Data Science AI Innovation Research Promotion Center
Shiga University
1-1-1, Bamba, Hikone, Shiga, 522-8522 Japan
Telephone: +81-749-27-1023
E-mail: bougtoir@gmail.com
ORCID: 0000-0001-7261-9062

19 May 2026

The Editors
Economica

Dear Professors Besley, den Haan, Ghatak, Ngai, Overman, and Thomas,

I am pleased to submit for your consideration the enclosed manuscript, **"What the Solow Residual Has Been Hiding: Tempo Drift and the Missing Intangible Share in National Capital Stocks"**, as an Article for *Economica*.

The paper addresses a fundamental question in growth accounting: how much of what we call TFP is a genuine productivity residual, and how much is a book-keeping artefact of mis-timed capital stocks and omitted intangibles? I show that two parameters — a time-varying investment-to-output lag μ(t) and an intangible-capital share β — jointly account for a measurable share of measured TFP variation across 39 OECD and middle-income economies (Penn World Table 10.01, World Bank CWON, 1995–2019). Letting μ drift reduces out-of-sample GDP forecast error by 13 %, and a Solow-residual decomposition reveals that up to 30 % of TFP-growth variance can be re-attributed to tempo drift and missing intangibles rather than genuine innovation. When both corrections are applied, production-side (flow) and wealth-side (stock) national accounts agree to within 1–2 % for most countries — the first empirical success, to my knowledge, of the "unified national-wealth accounting" programme that Stiglitz, Sen, and Fitoussi (2009) called for.

I believe this work is well suited to *Economica* for three reasons. First, it speaks directly to the economic substance of the Solow residual — a topic of central interest in macroeconomics and growth theory since Griliches (1996). The paper is framed not as a measurement exercise but as a discovery about what TFP has been measuring. Second, the identification exploits a structural analogy with the demographic tempo literature (Bongaarts and Feeney, 1998; Goldstein, Lutz, and Scherbov, 2003) that is novel in economics and generates falsifiable predictions. Third, the counterfactual wealth calculations have concrete policy implications for the Beyond-GDP debate, showing that official statistics systematically understate the productive capital base of innovation-intensive economies.

The manuscript is approximately 10,000 words of body text, with eleven figures and five tables.

I confirm the following:

- This work is original and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All data sources (Penn World Table 10.01; World Bank CWON; World Bank WDI) are publicly available; analysis code and intermediate outputs are archived in the public companion repository.
- I am the sole author.
- I have no competing interests, financial or otherwise, related to this work.
- I used generative AI to assist with formatting the text, choosing words, and writing analysis code; this is documented in the manuscript.

I would be grateful for your consideration.

Yours sincerely,

Onishi Tatsuki

Data Science AI Innovation Research Promotion Center

Shiga University

bougtoir@gmail.com